MITRC

MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

Name of Faculty: Dr. Raish Muhammad

Subject & Subject Code: Engineering Mathematics-2, 2FY1-01

Semester: II **Session: 2022-23 (Even Sem.)**

Branch: All

Unit No.: I **Date of Submission: 15/03/2023**

Lecture No.	Topics Covered	Page No.	Total Pages
1	**Matrices:** Rank of a matrix, Elementary transformations, Echelon form	1-4	4
2	Rank of matrix by reducing to normal forms	5-7	3
3	Consistency of system of linear simultaneous equations	8-11	4
4	Solution of system of linear equations	12-15	4
5	Eigenvalues and eigenvectors	16-19	4
6	Examples of Eigenvalues and eigenvectors	20-22	3
7	Diagonalization of matrix	23-26	4
8	Problems of diagonalization of matrices	27-30	4
9	Cayley-Hamilton theorem (without proof)	31-33	3
10	Examples of Cayley-Hamilton theorem	34-36	3

References:

1. Engineering Mathematics – D.C. Gokhroo & others, Unique Publications
2. Advanced Engineering Mathematics – Erwin Kreyszig, John Wiley & Sons
3. Advanced Engineering Mathematics – R.K. Jain & S.R.K. Iyengar, Narosa Publications
4. Higher Engineering Mathematics – B.V. Ramana, McGraw Hill Education
5. Advanced Engineering Mathematics – H.K. Das, S. Chand Pub.
6. Engineering Mathematics – Srimanta Pal & Subodh C. Bhunia, Oxford University Press
7. Engineering Mathematics – Y.N. Gaur & C.L. Koul, Jaipur Publishing House

Faculty **HOD** **Dean Academics**

UNIT-1 (MATRICES)

LECTURE NO.:- 1

Rank of a Matrix

A matrix A is said to be of rank r, if at least one of its minors (sub-determinants) of order r is non-zero and all minors of order greater than r are zero. It is written as rank (A) = r or $\rho(A) = r$.

The rank of matrices is used in the solution of simultaneous linear equations.

Properties of Rank of the Matrices

1. $\rho(A) = \rho(A^T)$.
2. $\rho(AB) \leq \min. \{\rho(A), \rho(B)\}$.
3. $\rho(O) = 0$.
4. If $A \neq O$, then $\rho(A) \geq 1$.
5. For a rectangular matrix A of order m×n, $\rho(A) \leq \min. \{m, n\}$.
   So if $A \neq O$, then $1 \leq \rho(A) \leq \min. \{m, n\}$.
6. For a square matrix A of order n, $\rho(A) \leq n$.
   So if $\rho(A) = n$, then $|A| \neq 0$, i.e. the matrix A is a non-singular matrix and if $\rho(A) < n$, then $|A| = 0$, i.e., the matrix A is a singular matrix.
7. If A and B are equivalent matrices, then $\rho(A) = \rho(B)$.
8. If A be a matrix of order m×n, then the nullity of the matrix A is
   $N(A) = n - \rho(A)$.

Example 1. Find rank of the following matrices:

(i) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$, (ii) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}$.

Solution (i). Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$.

Minor of order 3 of matrix A is $|A| = 0$. So $\rho(A) < 3$.

A minor of order 2 is $\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1 - 4 = -3 \neq 0$.

So $\rho(A) = 2$.

Hence the rank of the given matrix is 2. Ans.

(ii). Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \end{bmatrix}$.

Minors of order 2 are $\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 4 - 4 = 0$, $\begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = 5 - 6 = -1 \neq 0$ etc.

So $\rho(A) = 2$.

Hence the rank of the given matrix is 2. Ans.

Echelon Form of a Matrix

A matrix A is said to be in echelon form, if

(i) **zero row (s)** of A, which has all its elements zero **occur (s) below** a **non-zero row (s)**, which has non-zero element(s) and

(ii) the number of zeros before the first non-zero element in a row is less than the number of such zeros in the next row.

The rank of the matrix in echelon form is equal to the number of non-zero rows in the matrix.

Note. Use only row operations R_{ij}, kR_i, $R_i + kR_j$
or $R_i \leftrightarrow R_j$, $R_i \to kR_i$, $R_i \to R_i + kR_j$ $(k \neq 0)$.

Example 2. Determine the rank of the following matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}.$$

Solution. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$.

Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$, we get

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -2 & -4 \end{bmatrix}.$$

Now applying $R_3 \to R_3 - 2R_2$, we get

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}.$$ This is in echelon form having 2 non-zero rows.

So $\rho(A)$ = rank of echelon form = number of non-zero rows = 2.

Hence the rank of the given matrix is 2. Ans.

Example 3. Determine the rank of the following matrix:

$$\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}.$$

Solution. Let $A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}$.

Applying $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - R_1$, we get

$$A \sim \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & -6 \\ 0 & 0 & 0 & -2 \end{bmatrix}.$$

Now applying $R_3 \to R_3 - \frac{1}{3}R_2$, we get

$$A \sim \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$ This is in echelon form having 2 non-zero rows.

So $\rho(A)$ = rank of echelon form = number of non-zero rows = 2.

Hence the rank of the given matrix is 2. Ans.

Example 4. Determine rank of the matrix $A = \begin{bmatrix} -1 & 2 & 3 & -2 \\ 2 & -5 & 1 & 2 \\ 3 & -8 & 5 & 2 \\ 5 & -12 & -1 & 6 \end{bmatrix}$.

Solution. Given $A = \begin{bmatrix} -1 & 2 & 3 & -2 \\ 2 & -5 & 1 & 2 \\ 3 & -8 & 5 & 2 \\ 5 & -12 & -1 & 6 \end{bmatrix}$.

Applying $R_2 \to R_2 + 2R_1$, $R_3 \to R_3 + 3R_1$ and $R_4 \to R_4 + 5R_1$, we get

$$A \sim \begin{bmatrix} -1 & 2 & 3 & -2 \\ 0 & -1 & 7 & -2 \\ 0 & -2 & 14 & -4 \\ 0 & -2 & 14 & -4 \end{bmatrix}.$$

Now applying $R_3 \to R_3 - 2R_2$ and $R_4 \to R_4 - 2R_2$, we get

$$A \sim \begin{bmatrix} -1 & 2 & 3 & -2 \\ 0 & -1 & 7 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \text{ which is echelon form.}$$

We observe that the number of non-zero rows in the echelon form is 2.

Hence the rank of the given matrix A is 2. Ans.

Example 5. Find the rank of the following matrices:

$$\text{(i)} \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}, \quad \text{(ii)} \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}.$$

Solution. Try yourself.

Example 6. Compute rank of the following matrices using echelon form:

$$\text{(i)} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}, \quad \text{(ii)} \begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix}, \quad \text{(iii)} \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{bmatrix},$$

$$\text{(iv)} \begin{bmatrix} 1 & 2 & 1 & 0 \\ -2 & 4 & 3 & 0 \\ 1 & 0 & 2 & -8 \end{bmatrix}, \quad \text{(v)} \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}, \quad \text{(vi)} \begin{bmatrix} 2 & 4 & 3 & -2 \\ -3 & -2 & -1 & 4 \\ 6 & -1 & 7 & 2 \end{bmatrix},$$

$$\text{(vii)} \begin{bmatrix} 3 & 4 & 1 & 1 \\ 2 & 4 & 3 & 6 \\ -1 & -2 & 6 & 4 \\ 1 & -1 & 2 & -3 \end{bmatrix}, \quad \text{(viii)} \begin{bmatrix} 9 & 3 & 1 & 0 \\ 3 & 0 & 1 & -6 \\ 1 & 1 & 1 & 1 \\ 0 & -6 & 1 & 9 \end{bmatrix}.$$

Solution. Try yourself.

LECTURE NO.:- 2

Normal Form of a Matrix

If A be a matrix of rank r (> 0) and order m×n, then by applying elementary transformations (row operations and column operations), A can be reduced into one of the following forms:

$[I_r]$ $(m = n = r)$,

$[I_r \quad O]$ $(m = r, n > r)$,

$\begin{bmatrix} I_r \\ O \end{bmatrix}$ $(m > r, n = r)$,

$\begin{bmatrix} I_r & O \\ O & O \end{bmatrix}$ $(m > r, n > r)$.

These forms are called normal forms or canonical forms of the matrix A.

A zero matrix has its own normal form.

Example 1. Reduce the matrix $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ in its normal form and hence find its rank.

Solution. Given $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$.

Applying $R_1 \leftrightarrow R_2$, we get

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}.$$

Now applying $R_3 \to R_3 - 3R_1$ and $R_4 \to R_4 - R_1$, we get

$$A \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix}.$$

Applying $C_3 \to C_3 - C_1$ and $C_4 \to C_4 - C_1$, we get

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & -3 & -1 \end{bmatrix}.$$

Again applying $R_3 \to R_3 - R_2$ and $R_4 \to R_4 - R_2$, we get

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Further applying $C_3 \to C_3 + 3C_2$ and $C_4 \to C_4 + C_2$, we get

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_2 & O \\ O & O \end{bmatrix}.$$

This is the normal form of the matrix A. Ans.

Here r = 2. So $\rho(A) = 2$.

Hence the rank of the given matrix A is 2. Ans.

Example 2. Reduce the matrix $\begin{bmatrix} 1 & -1 & 3 & 6 \\ 1 & 3 & -3 & -4 \\ 2 & -2 & 6 & 12 \\ 5 & 3 & 3 & 11 \end{bmatrix}$ in its normal form and hence find its rank.

Solution. Let $A = \begin{bmatrix} 1 & -1 & 3 & 6 \\ 1 & 3 & -3 & -4 \\ 2 & -2 & 6 & 12 \\ 5 & 3 & 3 & 11 \end{bmatrix}$.

Try as above example 1. It is reduced in the following normal form:

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_3 & O \\ O & O \end{bmatrix}. \quad \text{Ans.}$$

Hence the rank of the given matrix is 3. Ans.

Example 3. Find rank of the matrix $\begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$ by reducing it into canonical form.

Solution. Let A = $\begin{bmatrix} 8 & 1 & 3 & 6 \\ 0 & 3 & 2 & 2 \\ -8 & -1 & -3 & 4 \end{bmatrix}$.

Try as above example 1. It is reduced in the following canonical form:

$$A \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = [I_3 \quad O].$$

Hence the rank of the given matrix is 3. Ans.

Example 4. Find the rank of the following matrices by reducing into normal form:

(i) $\begin{bmatrix} 3 & 2 & 5 & 7 & 12 \\ 1 & 1 & 2 & 3 & 5 \\ 3 & 3 & 6 & 9 & 15 \end{bmatrix}$, (ii) $\begin{bmatrix} 2 & -4 & 3 & 1 & 0 \\ 1 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$.

Solution. Try yourself.

Note: Nullity of a Matrix

The nullity of a matrix A is the dimension of the null space of A and is denoted by N(A).

The nullity of the identity matrix I is 0 and the nullity of the null matrix O of order m×n is n.

Rank-Nullity Theorem

The sum of the rank and the nullity of a matrix equals to the number of the columns of the matrix.

If A be a matrix of order m×n, then

$\rho(A) + N(A) = n.$

This theorem is also known as the dimension theorem.

LECTURE NO.:- 3

System of Linear Equations

A system of m linear equations in n unknowns $x_1, x_2, \cdots, x_n$ is of the form:

$a_{11}\, x_1 + a_{12}\, x_2 + \cdots + a_{1n}\, x_n = b_1,$

$a_{21}\, x_1 + a_{22}\, x_2 + \cdots + a_{2n}\, x_n = b_2,$

$\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\,,$

$a_{m1}\, x_1 + a_{m2}\, x_2 + \cdots + a_{mn}\, x_n = b_m.$

This system may be written in the following matrix form:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots (1)$$

$\Rightarrow AX = B,$

where $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$.

Non-Homogeneous System of Linear Equations

If at least one b_i (i = 1, 2, $\cdots$, m) is non-zero, then the above system (1) is called a non-homogeneous system of linear equations.

Homogeneous System of Linear Equations

If all b_i's (i = 1, 2, $\cdots$, m) are zero, then the above system (1) is called a homogeneous system of linear equations.

Solution of System of Linear Equations

A set of values of unknown variables $x_1, x_2, \cdots, x_n$, which satisfy all equations is called a solution of system of linear equations.

Consistency and Inconsistency of System of Linear Equations

A system is said to be consistent, if it has at least one set of solution. On the other hand, a system is said to be inconsistent, if it has no solution.

Matrix Method for Solution of System of Linear Equations

A system of linear equations may be written as AX = B,

where $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$.

This system may be expressed by an augmented matrix C = [A|B].

$$\Rightarrow C = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}.$$

Applying row operations, above matrix C may be reduced into echelon form and ranks of matrices A and C can be observed simultaneously. Here one of the following cases may be:

(i) $\rho(A) = \rho(C) = n$: The system is consistent and has unique solution.

(ii) $\rho(A) = \rho(C) < n$: The system is consistent and has infinite many solutions.

(iii) $\rho(A) \neq \rho(C)$: The system is inconsistent and has no solution.

If $\rho(A) = r$, then in case (i), r (= n) equations are obtained from echelon form and by back substitution, unique solution is obtained; in case (ii), r (< n) equations are obtained from echelon form and by back substitution, infinite many solutions are obtained in terms of n – r arbitrary values.

When system is **homogeneous**, B = O so that C = [A|O]. So $\rho(A) = \rho(C)$ always and hence a homogeneous system of linear equations is always consistent. Thus above case (iii) is not possible and there is no need to take augmented matrix C. Therefore one of the following two cases may be:

(i) $\rho(A) = n$: The homogeneous system has unique trivial (zero) solution.

(ii) $\rho(A) < n$: The homogeneous system has infinite many solutions containing one trivial (zero) solution and infinite many non-trivial (non-zero) solutions.

Example 1. Test the consistency, for the following system of equations and if possible, solve them:

$x + y + z = 6,\ x - y + 2z = 5,\ 3x + y + z = 8,\ 2x - 2y + 3z = 7.$

Solution. Given system of equations:

$x + y + z = 6,\ x - y + 2z = 5,\ 3x + y + z = 8,\ 2x - 2y + 3z = 7.$

Above system may be written as AX = B,

where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \\ 2 & -2 & 3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 6 \\ 5 \\ 8 \\ 7 \end{bmatrix}$.

Let us take an augmented matrix C = [A|B].

$$\Rightarrow C = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \\ 3 & 1 & 1 & 8 \\ 2 & -2 & 3 & 7 \end{bmatrix}.$$

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - 3R_1$ and $R_4 \to R_4 - 2R_1$, we get

$$C \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & -2 & -2 & -10 \\ 0 & -4 & 1 & -5 \end{bmatrix}.$$

Now applying $R_3 \to R_3 - R_2$ and $R_4 \to R_4 - 2R_2$, we get

$$C \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & -3 & -9 \\ 0 & 0 & -1 & -3 \end{bmatrix}.$$

Performing $R_3 \to (-\tfrac{1}{3})R_3$, we get

$$C \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & -1 & -3 \end{bmatrix}.$$

Again performing $R_4 \to R_4 + R_3$, we get

$$C \sim \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad \cdots\cdots\cdots\cdots\cdots \ (1)$$

From above echelon form, it is observed that

$\rho(A) = \rho(C) = 3$ = number of unknown variables.

So the given system is consistent and has unique solution. Ans.

Now from (1), we have

$$x + y + z = 6,$$
$$-2y + z = -1,$$
$$z = 3.$$

By back substitution, we have $z = 3$, $y = 2$, $x = 1$.

Hence the solution of the given system of equations is

$x = 1, y = 2, z = 3.$ Ans.

Example 2. Check the consistency and if possible, solve the system of equations:

$2x + 3y + 4z = 11$, $x + 5y + 7z = 15$, $3x + 11y + 13z = 25$.

Solution. Try as above example 1.

Example 3. Test the consistency, for the following system of equations and if possible, solve them:

$5x + 3y + 7z = 4$, $3x + 26y + 2z = 9$, $7x + 2y + 10z = 5$.

Solution. Try yourself.

LECTURE NO.:- 4

Examples of System of Linear Equations

Example 4. Check the consistency and if possible, solve the system of equations:

$4x - 5y + z = 2,\ 3x + y - 2z = 9,\ x + 4y + z = 5.$

Solution. Given system of equations is

$4x - 5y + z = 2,\ 3x + y - 2z = 9,\ x + 4y + z = 5.$

Above system may be written as AX = B,

where $A = \begin{bmatrix} 4 & -5 & 1 \\ 3 & 1 & -2 \\ 1 & 4 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ 9 \\ 5 \end{bmatrix}$.

Let us take an augmented matrix C = [A|B].

$$\Rightarrow C = \begin{bmatrix} 4 & -5 & 1 & 2 \\ 3 & 1 & -2 & 9 \\ 1 & 4 & 1 & 5 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_3$, we get

$$C \sim \begin{bmatrix} 1 & 4 & 1 & 5 \\ 3 & 1 & -2 & 9 \\ 4 & -5 & 1 & 2 \end{bmatrix}.$$

Now applying $R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 4R_1$, we get

$$C \sim \begin{bmatrix} 1 & 4 & 1 & 5 \\ 0 & -11 & -5 & -6 \\ 0 & -21 & -3 & -18 \end{bmatrix}.$$

Again performing $R_3 \to R_3 - \left(\frac{21}{11}\right)R_2$, we get

$$C \sim \begin{bmatrix} 1 & 4 & 1 & 5 \\ 0 & -11 & -5 & -6 \\ 0 & 0 & 72/11 & -72/11 \end{bmatrix}. \quad \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots (1)$$

From above echelon form, it is observed that

$\rho(A) = \rho(C) = 3$ = number of unknown variables.

So the given system is consistent and has unique solution. Ans.

Now from (1), we have

$x + 4y + z = 5,$

$-11y - 5z = -6,$

$(72/11)z = -72/11.$

By back substitution, we have $z = -1, y = 1, x = 2$.
Hence the solution of the given system of equations is
$x = 2, y = 1, z = -1$. Ans.

Example 5. Examine for consistency the following equations and solve them, if they are consistent:
$x + y + z = 6,\ 2x + y + 3z = 13,\ 5x + 2y + z = 12,\ 2x - 3y - 2z = -10$.

Solution. Try as above example 1.
Example 6. For what values of k, the equations
$x + y + z = 1,\ 2x + y + 4z = k,\ 4x + y + 10z = k^2$
have a solution and solve them completely in each case.

Solution. Given equations are
$x + y + z = 1,\ 2x + y + 4z = k,\ 4x + y + 10z = k^2$.

Above system may be written as AX = B,

where $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 4 \\ 4 & 1 & 10 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ k \\ k^2 \end{bmatrix}$.

Let us take an augmented matrix C = [A|B].

$$\Rightarrow C = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 4 & k \\ 4 & 1 & 10 & k^2 \end{bmatrix}.$$

Applying $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 4R_1$, we get

$$C \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & k-2 \\ 0 & -3 & 6 & k^2-4 \end{bmatrix}.$$

Again performing $R_3 \to R_3 - 3R_2$, we get

$$C \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & k-2 \\ 0 & 0 & 0 & k^2-3k+2 \end{bmatrix}. \quad \cdots\cdots\cdots (1)$$

From above echelon form, it is observed that the given system of equations will have a solution, when
$\rho(A) = \rho(C) = 2 < 3$ (number of unknown variables).
It is possible only when $k^2 - 3k + 2 = 0$ and then infinite many solutions exist.

$\Rightarrow (k - 1)(k - 2) = 0$

$\Rightarrow$ k = 1, 2. Ans.

Now from (1), we have

$x + y + z = 1,$

$-y + 2z = k - 2.$

For k = 1, we have

$x + y + z = 1, -y + 2z = -1.$

Let $z = a$, an arbitrary constant. Then we have

$x + y = 1 - a, -y = -2a - 1.$

By back substitution, we have $z = a$, $y = 2a + 1$, $x = -3a$.

Hence the solution of the given system of equations is

$x = -3a$, $y = 2a + 1$, $z = a$, where a is an arbitrary constant. Ans.

For k = 2, we have

$x + y + z = 1, -y + 2z = 0.$

Let $z = b$, an arbitrary constant. Then we have

$x + y = 1 - b, -y = -2b.$

By back substitution, we have $z = b$, $y = 2b$, $x = 1 - 3b$.

Hence the solution of the given system of equations is

$x = 1 - 3b$, $y = 2b$, $z = b$, where b is an arbitrary constant. Ans.

Example 7. Determine for what values of λ and μ, the following equations have (i) no solution, (ii) a unique solution and (iii) infinite number of solutions:

$x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu.$

Solution. Try yourself.

Example 8. Solve, if the following equations are consistent:

$2x + 6y + 11 = 0, 6x + 20y - 6z + 3 = 0, 6y - 18z + 1 = 0.$

Solution. Try yourself.

Example 9. Solve the following system of equations:

$x + y - 2z + 3w = 0, x - 2y + z - w = 0,$

$4x + y - 5z + 8w = 0, 5x - 7y + 2z - w = 0.$

Solution. Given system of equations is

$x + y - 2z + 3w = 0, x - 2y + z - w = 0,$

$4x + y - 5z + 8w = 0, 5x - 7y + 2z - w = 0.$

Above system is homogeneous and it may be written as AX = O, where $A = \begin{bmatrix} 1 & 1 & -2 & 3 \\ 1 & -2 & 1 & -1 \\ 4 & 1 & -5 & 8 \\ 5 & -7 & 2 & -1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$ and $O = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

Here we will reduce the coefficient matrix A into echelon form.

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - 4R_1$ and $R_4 \to R_4 - 5R_1$, we get

$$A \sim \begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & -3 & 3 & -4 \\ 0 & -12 & 12 & -16 \end{bmatrix}.$$

Now performing $R_3 \to R_3 - R_2$ and $R_4 \to R_4 - 4R_2$, we get

$$A \sim \begin{bmatrix} 1 & 1 & -2 & 3 \\ 0 & -3 & 3 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad \cdots\cdots\cdots\cdots (1)$$

From above echelon form, it is observed that

$\rho(A) = 2 < 4$ (number of unknown variables).

So the given system is consistent and has infinite many solutions.

Now from (1), we have

$x + y - 2z + 3w = 0,$

$-3y + 3z - 4w = 0.$

Let $z = a$ and $w = b$, where a and b are arbitrary constants. Then we have

$x + y = 2a - 3b$, $-3y = -3a + 4b$.

By back substitution, we have $y = a - 4b/3$, $x = a - 5b/3$.

Hence the solution of the given system of equations is

$x = a - 5b/3$, $y = a - 4b/3$, $z = a$, $w = b$, where a & b are arbitrary values. Ans.

Example 10. For which value of k, the following system of homogeneous equations has non-trivial solutions?

$2x + y + 2z = 0$, $x + y + 3z = 0$, $4x + 3y + kz = 0$.

Hint. For non-trivial solutions, rank of coefficient matrix < 3 (no. of variables).

$$\Rightarrow \begin{vmatrix} 2 & 1 & 2 \\ 1 & 1 & 3 \\ 4 & 3 & k \end{vmatrix} = 0 \Rightarrow k = 8. \quad \text{Ans.}$$

LECTURE NO.:- 5

Eigen Values and Eigen Vectors

If A be a square matrix of order n, I be the identity matrix of order n and λ be a scalar, then $A - \lambda I$ is called **characteristic matrix** of the matrix A; $|A - \lambda I|$ is a polynomial function of degree n in λ and is called **characteristic function or characteristic polynomial or eigen function** of the matrix A; $|A - \lambda I| = 0$ is called **characteristic equation** of the matrix A; roots of $|A - \lambda I| = 0$ are called **characteristic roots or latent roots or eigen values** of the matrix A and the set of the eigen values of the matrix A is called **spectrum** of the matrix A. Also if a non-zero vector X exists such that $AX = \lambda X$, then X is called **characteristic vector or latent vector or eigen vector** of the matrix A corresponding to the eigen value λ of the matrix A.

$AX = \lambda X \Rightarrow AX - \lambda X = O \Rightarrow AX - \lambda IX = O \Rightarrow (A - \lambda I)X = O$, which is a homogeneous system of linear equations and has infinite many non-trivial solutions to give parallel vectors, since $|A - \lambda I| = 0$.

Properties of Eigen Values and Eigen Vectors

1. Any square matrix of order n has n eigen values and the corresponding n linearly independent eigen vectors.
2. Any square matrix and its transpose have the same eigen values.
3. The sum of the eigen values of a matrix is equal to the trace of the matrix.
4. The product of the eigen values of a matrix is equal to its determinant.
5. If $\lambda_1, \lambda_2, \cdots, \lambda_n$ are the eigen values of a matrix A, then the eigen values of
   (i) kA are $k\lambda_1, k\lambda_2, \cdots, k\lambda_n$;
   (ii) A^m are $\lambda_1^m, \lambda_2^m, \cdots, \lambda_n^m$ and
   (iii) A^{-1} are $\lambda_1^{-1}, \lambda_2^{-1}, \cdots, \lambda_n^{-1}$.
6. A matrix is singular, if and only if one of its eigen values is zero.
7. The eigen vector of a matrix is not unique. If X is an eigen vector of a matrix A corresponding to the eigen value λ of the matrix A, then kX is also an eigen vector of the matrix A corresponding to the eigen value λ of the matrix A, where k is a non-zero scalar. So infinite many parallel eigen vectors of a matrix exist.
8. Two eigen vectors X_1 and X_2 are called orthogonal, if $X_1^T X_2 = 0 = X_2^T X_1$.

Example 1. Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}.$$

Solution. Given $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$.

The characteristic equation of the matrix A is

$|A - \lambda I| = 0.$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 0 & 1 \\ 0 & 2-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = 0$$

$\Rightarrow (2 - \lambda)\{(2 - \lambda)^2 - 0\} - 0 + 1\{0 - 1(2 - \lambda)\} = 0$

$\Rightarrow (2 - \lambda)\{ 4 - 4\lambda + \lambda^2 - 1\} = 0$

$\Rightarrow (2 - \lambda)\{ \lambda^2 - 4\lambda + 3\} = 0$

$\Rightarrow (2 - \lambda)\{ (\lambda - 1) (\lambda - 3)\} = 0.$

$\Rightarrow \lambda = 1, 2, 3.$

Now let $X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ and $X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$ be the eigen vectors of the given matrix A corresponding to the eigen values $\lambda = 1, 2$ and 3 respectively.

For $\lambda = 1$, by $(A - \lambda I)X = O$, we have

$$(A - 1I)X_1 = O \Rightarrow \begin{bmatrix} 2-1 & 0 & 1 \\ 0 & 2-1 & 0 \\ 1 & 0 & 2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_3 \rightarrow R_3 - R_1$, we get

$$\Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow x_1 + z_1 = 0, y_1 = 0.$

Let $x_1 = 1$. It gives $z_1 = -1$. Hence we get

$$X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

For $\lambda = 2$, by $(A - \lambda I)X = O$, we have

$$(A - 2I)X_2 = O \Rightarrow \begin{bmatrix} 2-2 & 0 & 1 \\ 0 & 2-2 & 0 \\ 1 & 0 & 2-2 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow z_2 = 0, x_2 = 0.$

Let $y_2 = 1$. Hence we get

$$X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

For $\lambda = 3$, by $(A - \lambda I)X = O$, we have

$$(A - 3I)X_3 = O \Rightarrow \begin{bmatrix} 2-3 & 0 & 1 \\ 0 & 2-3 & 0 \\ 1 & 0 & 2-3 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_3 \rightarrow R_3 + R_1$, we get

$$\Rightarrow \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow -x_3 + z_3 = 0, -y_3 = 0 \Rightarrow x_3 = z_3, y_3 = 0.$

Let $x_3 = 1$. It gives $z_3 = 1$. Hence we get

$$X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Thus the eigen values of the given matrix A are 1, 2 and 3 and the corresponding eigen vectors are $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Ans.

LECTURE NO.:- 6

Examples of Eigen Values and Eigen Vectors

Example 2. Find the eigen values and eigen vectors of the matrix

$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

Solution. Let $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$

The characteristic equation of the matrix A is

$|A - \lambda I| = 0.$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)\{(2-\lambda)^2 - 1\} + 1\{-1(2-\lambda) + 1\} + 1\{1 - 1(2-\lambda)\} = 0$$

$$\Rightarrow (2-\lambda)\{\,4 - 4\lambda + \lambda^2 - 1\} - 2 + \lambda + 1 + 1 - 2 + \lambda = 0$$

$$\Rightarrow (2-\lambda)\{\lambda^2 - 4\lambda + 3\} + 2\lambda - 2 = 0$$

$$\Rightarrow (2-\lambda)\{(\lambda-1)\,(\lambda-3)\} + 2(\lambda-1) = 0$$

$$\Rightarrow (\lambda-1)\{(2-\lambda)(\lambda-3) + 2\} = 0$$

$$\Rightarrow (\lambda-1)\{2\lambda - 6 - \lambda^2 + 3\lambda + 2\} = 0$$

$$\Rightarrow (\lambda-1)\{\lambda^2 - 5\lambda + 4\} = 0$$

$$\Rightarrow (\lambda-1)(\lambda-1)(\lambda-4) = 0$$

$$\Rightarrow \lambda = 1, 1, 4.$$

Now we shall find the eigen vectors of the given matrix A corresponding to the eigen values $\lambda = 1, 1$ and 4.

For $\lambda = 1, 1$, by $(A - \lambda I)X = O$, we have

$$(A - 1I)X = O \Rightarrow \begin{bmatrix} 2-1 & -1 & 1 \\ -1 & 2-1 & -1 \\ 1 & -1 & 2-1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_2 \to R_2 + R_1$ and $R_3 \to R_3 - R_1$, we get

$$\Rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow x - y + z = 0.$

If we choose $x = 1$, $y = 0$, then $z = -1$ and if we choose $x = 0$, $y = 1$, then $z = 1$.

$$\Rightarrow X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

For $\lambda = 4$, by $(A - \lambda I)X = O$, we have

$$(A - 4I)X = O \Rightarrow \begin{bmatrix} 2-4 & -1 & 1 \\ -1 & 2-4 & -1 \\ 1 & -1 & 2-4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & -1 & 1 \\ -1 & -2 & -1 \\ 1 & -1 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_3$, we get

$$\begin{bmatrix} 1 & -1 & -2 \\ -1 & -2 & -1 \\ -2 & -1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Now applying $R_2 \to R_2 + R_1$ and $R_3 \to R_3 + 2R_1$, we get

$$\begin{bmatrix} 1 & -1 & -2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Further applying $R_3 \to R_3 - R_2$, we get

$$\begin{bmatrix} 1 & -1 & -2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow x - y - 2z = 0, - 3y - 3z = 0.$

Let $z = 1$. It gives $y = -1$ and $x = -1 + 2 = 1$. Hence we get

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Thus the eigen values of the given matrix A are 1, 1 and 4 and the corresponding eigen vectors are $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. <u>Ans.</u>

Example 3. Find the eigen values and the corresponding eigen vectors of the matrix $A = \begin{bmatrix} -2 & 1 & 1 \\ -11 & 4 & 5 \\ -1 & 1 & 0 \end{bmatrix}$.

Solution. Try as above example 1.

Example 4. Find the eigen values and eigen vectors of the matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Solution. Try as above example 2.

Example 5. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$, then find the eigen values of $3A^3 + 5A^2 - 6A + 2I$.

Solution. Try yourself.

Example 6. If $\lambda_1, \lambda_2, \cdots, \lambda_n$ are the eigen values of a matrix A, then find the eigen values of the matrix $(A - \lambda I)^2$.

Solution. Try yourself.

Example 7. Show that the eigen values of a triangular matrix are just the diagonal elements of the matrix.

Solution. Try yourself.

Example 8. Prove that if λ is an eigen value of an orthogonal matrix, then $1/\lambda$ is also an eigen value of the matrix.

Solution. Try yourself.

LECTURE NO.:- 7

Diagonalization of a Matrix

Theorem. If a square matrix A of order n has n linearly independent eigen vectors $X_1, X_2, \cdots, X_n$ corresponding to the eigen values $\lambda_1, \lambda_2, \cdots, \lambda_n$, then the matrix A can be reduced to a diagonal form such that $P^{-1}AP = D$,
where $P = [X_1 \quad X_2 \quad \cdots \quad X_n]$ and $D = \text{diag.}(\lambda_1, \lambda_2, \cdots, \lambda_n)$.

Proof. We have

$$\begin{aligned} AP &= A[X_1 \quad X_2 \quad \cdots \quad X_n] \\ &= [AX_1 \quad AX_2 \quad \cdots \quad AX_n] \\ &= [\lambda_1 X_1 \quad \lambda_2 X_2 \quad \cdots \quad \lambda_n X_n] \\ &= [X_1 \quad X_2 \quad \cdots \quad X_n]\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \\ &= PD. \end{aligned}$$

Pre-multiplying both sides of above equation by P^{-1}, we get
$P^{-1}AP = P^{-1}PD = ID = D$.

$\Rightarrow P^{-1}AP = D$. Proved.

Remark. The transformation of a matrix A to $P^{-1}AP = D$ is known as a **similarity transformation**. Here the matrix P is known as the **modal matrix** and D is known as the **spectral matrix** of the matrix A.

Powers of a Matrix

We have $D = P^{-1}AP$.

Then $D^2 = (P^{-1}AP)(P^{-1}AP) = P^{-1}A(PP^{-1})AP$
$= P^{-1}AIAP = P^{-1}AAP$
$= P^{-1}A^2P$.

Similarly $D^3 = P^{-1}A^3P$.

Continuing in this way, we have $D^m = P^{-1}A^mP$.

Pre-multiplying by P and post-multiplying by P^{-1}, we get

$PD^mP^{-1} = P(P^{-1}A^mP)P^{-1}$
$= (PP^{-1})A^m(PP^{-1}) = IA^mI$
$= A^n$.

$\Rightarrow A^m = PD^mP^{-1}$.

Example 1. Find the eigen values and corresponding eigen vectors for the matrix

$$A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}.$$

Also reduce it to the diagonal form. Obtain the modal matrix.

Solution. Given $A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$.

The characteristic equation of the matrix A is

$|A - \lambda I| = 0$.

$$\Rightarrow \begin{vmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & 0-\lambda \end{vmatrix} = 0$$

$\Rightarrow (-1-\lambda)\{(2-\lambda)(-\lambda)+1\} - 2\{-\lambda+1\} - 2\{-1+1(2-\lambda)\} = 0$

$\Rightarrow -(1+\lambda)\{-2\lambda+\lambda^2+1\} - 2\{1-\lambda\} - 2\{1-\lambda\} = 0$

$\Rightarrow -(\lambda+1)\{(\lambda-1)^2\} + 2(\lambda-1) + 2(\lambda-1) = 0$

$\Rightarrow (\lambda+1)(\lambda-1)^2 - 4(\lambda-1) = 0$

$\Rightarrow (\lambda-1)\{(\lambda+1)(\lambda-1) - 4\} = 0$

$\Rightarrow (\lambda-1)\{\lambda^2 - 1 - 4\} = 0 \Rightarrow (\lambda-1)\{\lambda^2 - 5\} = 0$

$\Rightarrow (\lambda-1)(\lambda-\sqrt{5})(\lambda+\sqrt{5}) = 0$

$\Rightarrow \lambda = 1, \sqrt{5}, -\sqrt{5}$.

Now let $X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ and $X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$ be the eigen vectors of the given matrix A corresponding to the eigen values $\lambda = 1, \sqrt{5}$ and $-\sqrt{5}$ respectively.

For $\lambda = 1$, by $(A - \lambda I)X = O$, we have

$$(A - 1I)X_1 = O \Rightarrow \begin{bmatrix} -1-1 & 2 & -2 \\ 1 & 2-1 & 1 \\ -1 & -1 & 0-1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & 2 & -2 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_2$, we get

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ -2 & 2 & -2 \\ -1 & -1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Again applying $R_2 \to R_2 + 2R_1$ and $R_3 \to R_3 + R_1$, we get

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow x_1 + y_1 + z_1 = 0,\ y_1 = 0.$

Let $x_1 = 1$. It gives $z_1 = -1$. Hence we get

$$X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

For $\lambda = \sqrt{5}$, by $(A - \lambda I)X = O$, we have

$$(A - \sqrt{5}I)X_2 = O \Rightarrow \begin{bmatrix} -1-\sqrt{5} & 2 & -2 \\ 1 & 2-\sqrt{5} & 1 \\ -1 & -1 & 0-\sqrt{5} \end{bmatrix}\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_2$, we get

$$\Rightarrow \begin{bmatrix} 1 & 2-\sqrt{5} & 1 \\ -1-\sqrt{5} & 2 & -2 \\ -1 & -1 & -\sqrt{5} \end{bmatrix}\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Again applying $R_2 \to R_2 + (1 + \sqrt{5})R_1$ and $R_3 \to R_3 + R_1$, we get

$$\Rightarrow \begin{bmatrix} 1 & 2-\sqrt{5} & 1 \\ 0 & \sqrt{5}-1 & \sqrt{5}-1 \\ 0 & 1-\sqrt{5} & 1-\sqrt{5} \end{bmatrix}\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Now applying $R_3 \to R_3 - \left(\frac{1-\sqrt{5}}{\sqrt{5}-1}\right)R_2$, we get

$$\Rightarrow \begin{bmatrix} 1 & 2-\sqrt{5} & 1 \\ 0 & \sqrt{5}-1 & \sqrt{5}-1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow x_2 + (2 - \sqrt{5})y_2 + z_2 = 0,\ (\sqrt{5} - 1)y_2 + (\sqrt{5} - 1)z_2 = 0.$

Let $z_2 = 1$. It gives $y_2 = -1$ and $x_2 = 1 - \sqrt{5}$. Hence we get

$$X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 1-\sqrt{5} \\ -1 \\ 1 \end{bmatrix}.$$

For $\lambda = -\sqrt{5}$, by $(A - \lambda I)X = O$, we have

$$(A + \sqrt{5}I)X_3 = O \Rightarrow \begin{bmatrix} -1+\sqrt{5} & 2 & -2 \\ 1 & 2+\sqrt{5} & 1 \\ -1 & -1 & 0+\sqrt{5} \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_2$, we get

$$\Rightarrow \begin{bmatrix} 1 & 2+\sqrt{5} & 1 \\ -1+\sqrt{5} & 2 & -2 \\ -1 & -1 & \sqrt{5} \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Again applying $R_2 \to R_2 + (1 - \sqrt{5})R_1$ and $R_3 \to R_3 + R_1$, we get

$$\Rightarrow \begin{bmatrix} 1 & 2+\sqrt{5} & 1 \\ 0 & -\sqrt{5}-1 & -\sqrt{5}-1 \\ 0 & 1+\sqrt{5} & 1+\sqrt{5} \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Now applying $R_3 \to R_3 + R_2$, we get

$$\Rightarrow \begin{bmatrix} 1 & 2+\sqrt{5} & 1 \\ 0 & -\sqrt{5}-1 & -\sqrt{5}-1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow x_3 + (2 + \sqrt{5})y_3 + z_3 = 0, (-\sqrt{5} - 1)y_3 + (-\sqrt{5} - 1)z_3 = 0.$

Let $z_3 = 1$. It gives $y_3 = -1$ and $x_3 = 1 + \sqrt{5}$. Hence we get

$$X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1+\sqrt{5} \\ -1 \\ 1 \end{bmatrix}.$$

Thus the eigen values of the given matrix A are 1, $\sqrt{5}$ and $-\sqrt{5}$ and the corresponding eigen vectors are $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 1-\sqrt{5} \\ -1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1+\sqrt{5} \\ -1 \\ 1 \end{bmatrix}$. Ans.

The given matrix A is now reduced to the diagonal form as

$$P^{-1}AP = D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & -\sqrt{5} \end{bmatrix},$$

where $P = \begin{bmatrix} 1 & 1-\sqrt{5} & 1+\sqrt{5} \\ 0 & -1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$ is the modal matrix. Ans.

LECTURE NO.:- 8

Examples on Diagonalization of Matrices

Example 2. Find the matrix, which transforms the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \text{ to a diagonal form.}$$

Solution. Given $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.

The characteristic equation of the matrix A is

$|A - \lambda I| = 0.$

$$\Rightarrow \begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0$$

$\Rightarrow (8 - \lambda)\{(7 - \lambda)(3 - \lambda) - 16\} + 6\{- 6(3 - \lambda) + 8\} + 2\{24 - 2(7 - \lambda)\} = 0$

$\Rightarrow (8 - \lambda)\{21 - 10\lambda + \lambda^2 - 16\} + 6\{- 18 + 6\lambda + 8\} + 2\{24 - 14 + 2\lambda\} = 0$

$\Rightarrow (8 - \lambda)\{\lambda^2 - 10\lambda + 5\} + 6\{6\lambda - 10\} + 2\{10 + 2\lambda\} = 0$

$\Rightarrow 8\lambda^2 - 80\lambda + 40 - \lambda^3 + 10\lambda^2 - 5\lambda + 36\lambda - 60 + 20 + 4\lambda = 0$

$\Rightarrow - \lambda^3 + 18\lambda^2 - 45\lambda = 0$

$\Rightarrow - \lambda(\lambda^2 - 18\lambda + 45) = 0$

$\Rightarrow \lambda(\lambda - 3)(\lambda - 15) = 0$

$\Rightarrow \lambda = 0, 3, 15.$

Now let $X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ and $X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$ be the eigen vectors of the given matrix A corresponding to the eigen values $\lambda = 0$, 3 and 15 respectively.

For $\lambda = 0$, by $(A - \lambda I)X = O$, we have

$(A - 0I)X_1 = O \Rightarrow AX_1 = O.$

$$\Rightarrow \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \leftrightarrow R_3$, we get

$$\begin{bmatrix} 2 & -4 & 3 \\ -6 & 7 & -4 \\ 8 & -6 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Again applying $R_2 \rightarrow R_2 + 3R_1$ and $R_3 \rightarrow R_3 - 4R_1$, we get

$$\begin{bmatrix} 2 & -4 & 3 \\ 0 & -5 & 5 \\ 0 & 10 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Now applying $R_3 \rightarrow R_3 + 2R_2$, we get

$$\begin{bmatrix} 2 & -4 & 3 \\ 0 & -5 & 5 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow 2x_1 - 4y_1 + 3z_1 = 0, -5y_1 + 5z_1 = 0.$

$\Rightarrow 2x_1 - y_1 = 0, y_1 = z_1.$

Let $x_1 = 1$. It gives $y_1 = z_1 = 2$. Hence we get

$$X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

For $\lambda = 3$, by $(A - \lambda I)X = O$, we have

$$(A - 3I)X_2 = O \Rightarrow \begin{bmatrix} 8-3 & -6 & 2 \\ -6 & 7-3 & -4 \\ 2 & -4 & 3-3 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \rightarrow R_1 + R_2$, we get

$$\begin{bmatrix} -1 & -2 & -2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Again applying $R_2 \rightarrow R_2 - 6R_1$ and $R_3 \rightarrow R_3 + 2R_1$, we get

$$\begin{bmatrix} -1 & -2 & -2 \\ 0 & 16 & 8 \\ 0 & -8 & -4 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Now applying $R_3 \to R_3 + ½R_2$, we get

$$\begin{bmatrix} -1 & -2 & -2 \\ 0 & 16 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow -x_2 - 2y_2 - 2z_2 = 0,\ 16y_2 + 8z_2 = 0.$

Let $y_2 = 1$. It gives $z_2 = -2$ and $x_2 = 2$. Hence we get

$$X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}.$$

For $\lambda = 15$, by $(A - \lambda I)X = O$, we have

$$(A - 15I)X_3 = O \Rightarrow \begin{bmatrix} 8-15 & -6 & 2 \\ -6 & 7-15 & -4 \\ 2 & -4 & 3-15 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Applying $R_1 \to R_1 - R_2$, we get

$$\begin{bmatrix} -1 & 2 & 6 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Again applying $R_2 \to R_2 - 6R_1$ and $R_3 \to R_3 + 2R_1$, we get

$$\begin{bmatrix} -1 & -8 & 10 \\ 0 & -20 & -40 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$\Rightarrow -x_3 - 8y_3 + 10z_3 = 0,\ -20y_3 - 40z_3 = 0.$

Let $z_3 = 1$. It gives $y_3 = -2$ and $x_3 = 2$. Hence we get

$$X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}.$$

Thus the given matrix A is now reduced to the diagonal form as

$$P^{-1}AP = D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix},$$

where $P = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ is the modal matrix, which transforms the given matrix A to diagonal form. Ans.

Example 3. Find a similarity transformation, which will reduce the matrix

$$A = \begin{bmatrix} 2 & -3 & 3 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$ to a diagonal form.

Solution. Try as above example 1.

Example 4. The matrix $A = \begin{bmatrix} a & h \\ h & b \end{bmatrix}$ is transformed to the diagonal form

$D = T^{-1}AT$, where $T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$.

Find the value of θ, which gives this diagonal transformation.

Solution. Try yourself.

Example 5. A square matrix A is given by $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$.

Find the modal matrix P and resulting diagonal matrix D of A.

Hence calculate A^4.

Hint. $A^4 = P\,D^4\,P^{-1}$.

LECTURE NO.:- 9

Cayley-Hamilton Theorem

Statement: Every square matrix satisfies its own characteristic equation.

Explanation: If A be a square matrix of order n, then its characteristic equation is

$|A - \lambda I| = 0 \Rightarrow (-1)^n\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_{n-1}\lambda + a_n = 0.$

By Cayley-Hamilton theorem, matrix A satisfies above equation so that

$(-1)^n A^n + a_1 A^{n-1} + a_2 A^{n-2} + \cdots + a_{n-1}A + a_n I = O.$

Inversion by Cayley-Hamilton Theorem

Multiplying above equation by A^{-1}, we get

$(-1)^n A^{n-1} + a_1 A^{n-2} + a_2 A^{n-3} + \cdots + a_{n-1}I + a_n A^{-1} = O.$

$\Rightarrow (-1)^n A^{n-1} + a_1 A^{n-2} + a_2 A^{n-3} + \cdots + a_{n-1}I = O - a_n A^{-1} = - a_n A^{-1}.$

$\therefore\ A^{-1} = -\frac{1}{a_n}[(-1)^n A^{n-1} + a_1 A^{n-2} + a_2 A^{n-3} + \cdots + a_{n-1}I].$

Example 1. Verify Cayley-Hamilton theorem for the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}. \text{ Hence find } A^{-1}.$$

Solution. Given $A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$

The characteristic equation of the matrix A is

$|A - \lambda I| = 0.$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$\Rightarrow (2-\lambda)\{(2-\lambda)^2 - 1\} - 1\{-1(2-\lambda) + 1\} + 1\{1 - 1(2-\lambda)\} = 0$

$\Rightarrow (2-\lambda)\{4 - 4\lambda + \lambda^2 - 1\} + 2 - \lambda - 1 + 1 - 2 + \lambda = 0$

$\Rightarrow (2-\lambda)\{\lambda^2 - 4\lambda + 3\} = 0$

$\Rightarrow 2\lambda^2 - 8\lambda + 6 - \lambda^3 + 4\lambda^2 - 3\lambda = 0$

$\Rightarrow -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

$\Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0.$ ·············· (1)

According to Cayley-Hamilton theorem, matrix A satisfies above characteristic equation (1) so that

$A^3 - 6A^2 + 11A - 6I = O.$ ·············· (2)

Here we need to verify above equation (2). So we have to find A^2 and A^3.

$$\therefore\ A^2 = AA = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 3 & 3 \\ -5 & 4 & -5 \\ 5 & -3 & 6 \end{bmatrix}$$

$$\&\ A^3 = A^2A = \begin{bmatrix} 4 & 3 & 3 \\ -5 & 4 & -5 \\ 5 & -3 & 6 \end{bmatrix}\begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 7 & 7 \\ -19 & 8 & -19 \\ 19 & -7 & 20 \end{bmatrix}.$$

Putting the values of A^3, A^2, A and I, we have

L.H.S. of (2) = $A^3 - 6A^2 + 11A - 6I$

$$= \begin{bmatrix} 8 & 7 & 7 \\ -19 & 8 & -19 \\ 19 & -7 & 20 \end{bmatrix} - 6\begin{bmatrix} 4 & 3 & 3 \\ -5 & 4 & -5 \\ 5 & -3 & 6 \end{bmatrix} + 11\begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 6\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8-24+22-6 & 7-18+11-0 & 7-18+11-0 \\ -19+30-11-0 & 8-24+22-6 & -19+30-11-0 \\ 19-30+11-0 & -7+18-11-0 & 20-36+22-6 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

= O

= R.H.S. of (2).

So the given matrix A satisfies its characteristic equation (1).

Hence the Cayley-Hamilton theorem is <u>verified</u> for the given matrix A.

Now multiplying equation (2) by A^{-1}, we have

$A^2 - 6A + 11I - 6A^{-1} = O.$

$\Rightarrow 6A^{-1} = A^2 - 6A + 11I$

$$= \begin{bmatrix} 4 & 3 & 3 \\ -5 & 4 & -5 \\ 5 & -3 & 6 \end{bmatrix} - 6\begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 11\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4-12+11 & 3-6+0 & 3-6+0 \\ -5+6+0 & 4-12+11 & -5+6+0 \\ 5-6+0 & -3+6+0 & 6-12+11 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & -3 & -3 \\ 1 & 3 & 1 \\ -1 & 3 & 5 \end{bmatrix}.$$

Hence $A^{-1} = \frac{1}{6}\begin{bmatrix} 3 & -3 & -3 \\ 1 & 3 & 1 \\ -1 & 3 & 5 \end{bmatrix}$. <u>Ans.</u>

Example 2. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.

Show that the equation is satisfied by A and hence find A^{-1}.

Solution. Try yourself as above example 1.

Example 3. State Cayley-Hamilton theorem. Verify it for the matrix

$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$. Hence find A^{-1}.

Solution. Try yourself.

Example 4. Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

and verify that it is satisfied by A.

Solution. Try yourself.

LECTURE NO.:- 10

Examples on Cayley-Hamilton Theorem

Example 5. If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, then express

$A^6 - 6A^5 + 9A^4 - 2A^3 - 12A^2 + 23A - 9I$ as a linear polynomial in A.

Solution. Given $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.

The characteristic equation of the matrix A is
$|A - \lambda I| = 0.$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)\{(2-\lambda)^2 - 1\} + 1\{-1(2-\lambda) + 1\} + 1\{1 - 1(2-\lambda)\} = 0$$

$$\Rightarrow (2-\lambda)\{4 - 4\lambda + \lambda^2 - 1\} + \{-2 + \lambda + 1\} + \{1 - 2 + \lambda\} = 0$$

$$\Rightarrow (2-\lambda)\{\lambda^2 - 4\lambda + 3\} - 1 + \lambda - 1 + \lambda = 0$$

$$\Rightarrow 2\lambda^2 - 8\lambda + 6 - \lambda^3 + 4\lambda^2 - 3\lambda - 2 + 2\lambda = 0$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 - 9\lambda + 4 = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0. \cdots\cdots\cdots\cdots\cdots (1)$$

By Cayley-Hamilton theorem, above characteristic equation (1) is satisfied by the matrix A, then we have

$$A^3 - 6A^2 + 9A - 4I = O. \cdots\cdots\cdots\cdots\cdots (2)$$

Now we have

$A^6 - 6A^5 + 9A^4 - 2A^3 - 12A^2 + 23A - 9I$

$= A^3(A^3 - 6A^2 + 9A - 4I) + 2A^3 - 12A^2 + 23A - 9I$

$= A^3(O) + 2(A^3 - 6A^2 + 9A - 4I) + 5A - I$, by (2)

$= O + 2(O) + 5A – I$, by (2)

$= 5A – I$. Ans.

Example 6. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.

Show that the equation is satisfied by A and hence find A^{-1}.

Solution. Given $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.

The characteristic equation of the matrix A is

$|A – \lambda I| = 0$.

$$\Rightarrow \begin{vmatrix} 1-\lambda & 3 & 7 \\ 4 & 2-\lambda & 3 \\ 1 & 2 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1 – \lambda)\{(2 – \lambda)(1 – \lambda) – 6\} – 3\{4(1 – \lambda) – 3\} + 7\{8 – 1(2 – \lambda)\} = 0$$

$$\Rightarrow (1 – \lambda)\{ 2 – 3\lambda + \lambda^2 – 6\} – 3\{4 – 4\lambda – 3\} + 7\{8 – 2 + \lambda\} = 0$$

$$\Rightarrow (1 – \lambda)\{ \lambda^2 – 3\lambda – 4\} – 3\{1 – 4\lambda\} + 7\{6 + \lambda\} = 0$$

$$\Rightarrow \lambda^2 – 3\lambda – 4 – \lambda^3 + 3\lambda^2 + 4\lambda – 3 + 12\lambda + 42 + 7\lambda = 0$$

$$\Rightarrow – \lambda^3 + 4\lambda^2 + 20\lambda + 35 = 0. \quad \underline{\text{Ans.}}$$

$$\Rightarrow \lambda^3 – 4\lambda^2 – 20\lambda – 35 = 0. \cdots\cdots\cdots\cdots\cdots (1)$$

If the above characteristic equation (1) is satisfied by the matrix A, then

$$A^3 – 4A^2 – 20A – 35I = O. \cdots\cdots\cdots\cdots\cdots (2)$$

Here we need to verify above equation (2). So we have to find A^2 and A^3.

$$\therefore A^2 = AA = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{bmatrix}$$

$$\& A^3 = A^2A = \begin{bmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{bmatrix}\begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{bmatrix}.$$

Putting the values of A^3, A^2, A and I, we have

L.H.S. of (2) $= A^3 – 4A^2 – 20A – 35I$

$$= \begin{bmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{bmatrix} - 4\begin{bmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{bmatrix} - 20\begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} - 35\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 135-80-20-35 & 152-92-60-0 & 232-92-140-0 \\ 140-60-80-0 & 163-88-40-35 & 208-148-60-0 \\ 60-40-20-0 & 76-36-40-0 & 111-56-20-35 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$= O$

$=$ R.H.S. of (2).

Hence the characteristic equation of the given matrix A is satisfied by A.

Proved.

Now multiplying equation (2) by A^{-1}, we have

$A^2 - 4A - 20I - 35A^{-1} = O$.

$\Rightarrow 35A^{-1} = A^2 - 4A - 20I$

$$= \begin{bmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{bmatrix} - 4\begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix} - 20\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 20-4-20 & 23-12-0 & 23-28-0 \\ 15-16-0 & 22-8-20 & 37-12-0 \\ 10-4-0 & 9-8-0 & 14-4-20 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ 6 & 1 & -10 \end{bmatrix}.$$

Hence $A^{-1} = \frac{1}{35}\begin{bmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ 6 & 1 & -10 \end{bmatrix}$. Ans.